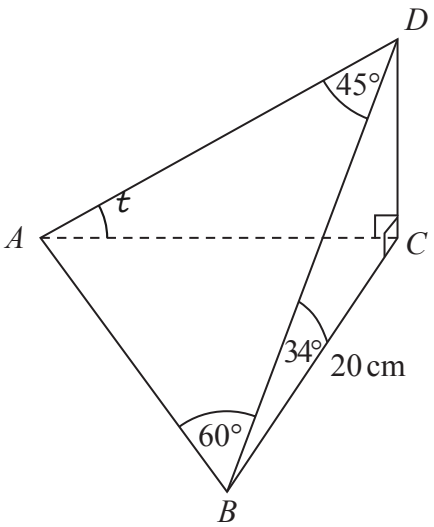
19 The diagram shows a pyramid with base ABC.



CD is perpendicular to both CA and CB.
Angle $CBD = 34°$ Angle $ADB = 45°$ Angle $DBA = 60°$
$BC = 20$ cm.

Calculate the size of the angle between the line AD and the plane ABC.
Give your answer correct to 1 decimal place.

The required angle is t = angle DAC: C is the vertical foot.

CD = 20 tan 34° = 13.49017034... cm.

1 mark
Tangent in BCD

BD = 20 ÷ cos 34° = 24.12435897... cm.

1 mark
Hypotenuse BD

In ADB, angle A = 180° − 45° − 60° = 75°.

AD = BD × sin 60° ÷ sin 75° = 21.62930853... cm.

1 mark
Sine rule in ADB

sin t = CD ÷ AD = 13.49017034... ÷ 21.62930853...

1 mark
Sine in ACD

t = 38.58673477...° ≈ 38.6°

1 mark
38.6° (1 d.p.)

°

(Total for Question 19 is 5 marks)